



# Mono-component multiparameter acoustic full waveform inversion in vertically transverse isotropic media using converted vector wavefields

Youngjae Shin <sup>a</sup>, Ju-Won Oh <sup>b,\*</sup>, Shinwoong Kim <sup>c</sup>, Dong-Joo Min <sup>a,d</sup>

<sup>a</sup> Seoul National University, Dept. of Energy Systems Engineering, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea

<sup>b</sup> Chonbuk National University, Dept. of Resources and Energy Engineering, 567, Baekje-daero, Deokjin-gu, Jeonju-si, Jeollabuk-do 54896, Republic of Korea

<sup>c</sup> SK Innovation, 26, Jong-ro, Jongro-gu, Seoul 03188, Republic of Korea

<sup>d</sup> Research institute of energy and resources, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea

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## ABSTRACT

In multiparameter full-waveform inversion (FWI) with acoustic-approximation for vertically transverse isotropic (VTI) media, it is important to choose an appropriate modeling technique for computational efficiency and numerical stability. In addition, because the gradient is determined by the modeling algorithm used in the FWI process, we need to examine if the modeling algorithm yields a gradient similar to that obtained from the elastic scattering theory. The decomposed equation with mono-component has been proposed because it is computationally efficient and does not generate shear-wave artifacts. However, the decomposed equation has the limitation that its numerical scattering potentials for the perturbations of anisotropic parameters do not properly simulate the theoretical elastic scattering potentials. To overcome the limitation, we propose using the vector virtual sources to compute the gradient in the FWI algorithm by converting the mono-component (pressure) source and receiver wavefields into the two-component particle displacements using the equation of motion on the staggered grids. Unlike the conventional method, our approach properly simulates PP radiation patterns for anisotropic parameters obtained under the elastic assumption, while retaining computational efficiency achieved by using the mono-component. Numerical examples show that anisotropic parameters are updated in the correct gradient direction by our method in the FWI process.

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## 1. Introduction

Full waveform inversion (FWI) has been highlighted as a promising technique to provide background velocity models for subsurface reflector imaging. When we apply seismic imaging techniques such as FWI and migration to seismic data acquired in sedimentary basins, we need to consider the anisotropic parameters in addition to seismic velocity because seismic anisotropy is commonly encountered in sedimentary basins. Therefore, the estimation of subsurface anisotropic properties plays an important role in locating subsurface reflectors in the right positions.

For the realistic physical assumption, elastic FWI based on elastic modeling has been developed and successfully applied for isotropic media. The main advantages of elastic FWI are that it provides reliable subsurface images and elastic properties such as density and the ratio of P-wave and S-wave velocities, which are essential for reservoir characterization (Pan et al., 2018a, 2019). However, it is still difficult to obtain good results with elastic FWI due to the complex mode

conversion, strong surface waves and trade-off issues. In addition, elastic FWI requires suffers high computational costs, particularly in 3D data inversion. For this reason, FWI for elastic media has been performed under the acoustic and isotropic assumptions. However, those assumptions cause significant errors, specifically for the long-offset data that FWI mainly relies on (i.e., diving wave data).

To consider anisotropic properties in FWI, the acoustic-approximation of elastic wave equations has been introduced in modeling and inversion algorithms. Many studies have been devoted to improving the accuracy and stability of the acoustic-based modeling and inversion algorithms for anisotropic media (Alkhalifah, 2000; Zhou et al., 2006; Duveneck et al., 2008; Song and Alkhalifah, 2013; Xu and Zhou, 2014; Xu et al., 2015). Alkhalifah (2000) introduced an acoustic-approximation for VTI media by setting the S-wave velocity along the symmetry axis to zero. By separating S-waves from the elastic wave propagation, we only simulate the propagation of P-waves, which enhances computational efficiency. Zhou et al. (2006) rearranged the fourth-order differential equation proposed by Alkhalifah (2000) by introducing the auxiliary variables to obtain a couple of second-order differential equations. Duveneck et al. (2008) presented another coupled system for pseudo-acoustic P-wave modeling by combining the linear stress-strain relationship and the equation of motion.

\* Corresponding author.

E-mail address: [juwon24@jbnu.ac.kr](mailto:juwon24@jbnu.ac.kr) (J.-W. Oh).

The advantage of solving the coupled systems compared to the original fourth-order wave equation is computational speedup, whereas their limitation is the numerical shear wave artifacts arising from introducing two wavefield components (Grechka et al., 2004; Duveneck et al., 2008). The coupled systems are also vulnerable when we describe severely tilted transversely isotropic (TTI) media.

On the other hand, Xu and Zhou (2014) decomposed the original pseudo-differential operator into the Laplacian and scalar operators. While the coupled systems are composed of two normal stress components, this decomposed equation consists of one differential equation with a single pressure component. The main advantages of the decomposed equation compared to the coupled system are that it does not generate the shear-wave artifacts and the computational cost is significantly reduced. Furthermore, the decomposed equation can be easily extended to TTI and rotated orthorhombic media while suppressing the shear wave artifacts in the symmetric axis. Xu et al. (2015) proposed the elliptical operator instead of the Laplacian operator, which gives more balanced amplitudes and reduces numerical errors for strongly anisotropic media.

In multiparameter anisotropic FWI, it is essential to understand the scattering potential of inversion parameters (so-called radiation pattern) to minimize the trade-off between parameters. For this purpose, many FWI algorithms using different acoustic-approximated modeling schemes have been presented for anisotropic media (Operto et al., 2013; Gholami et al., 2013a, 2013b; Le et al., 2015; Masmoudi and Alkhalifah, 2017). Based on the coupled system of pseudo-acoustic wave equations in a VTI media (Duveneck et al., 2008), Gholami et al. (2013a) analyzed the scattering potentials of anisotropic parameters from the different parameterizations and suggested an appropriate combination of anisotropic parameters for inversion considering the data acquisition and the quality of the initial model. Le et al. (2015) used a linear pseudo-differential equation (Xu and Zhou, 2014) and derived anisotropic FWI using first-order Born perturbation modeling to linearize the forward equation and the adjoint method. Masmoudi and Alkhalifah (2017) applied acoustic-approximation anisotropic modeling with the elliptic decomposition (Xu and Zhou, 2014) for acoustic orthorhombic FWI and analyzed the gradient with a practical set of parameters.

On the other hand, the scattering potentials for model parameters have been theoretically described from elastic scattering theory (Wu and Aki, 1988; Tarantola, 1986; Tsvankin, 1997; Alkhalifah and Plessix, 2014; Oh and Alkhalifah, 2016a; Kamath and Tsvankin, 2016; Pan et al., 2016; Kim and Min, 2014). In contrast, the numerical scattering potential of anisotropic parameters by the acoustic-approximation with a single component for anisotropic media (Xu and Zhou, 2014) has not been examined yet.

In this study, we first examine the numerical radiation patterns of the partial derivative wavefields for the anisotropic parameters using the decomposed acoustic wave equation. We then show that those numerical scattering patterns deviate from the theoretical expectations because the single component is used in the decomposed acoustic equation. To resolve this issue, we adopt elastic scattering theory to compute the gradient based on the wave equations expressed in the first-order form on the staggered grids (Virieux, 1986; Graves, 1996). Therefore, we can correctly describe the scattering potential of anisotropic parameters with the decomposed acoustic wave equation.

In the following sections, we review the basic theory of acoustic-approximation anisotropic modeling and FWI and reveal the limitation of the single-component wave equation by describing the gradient directions of anisotropic parameters based on the numerical radiation pattern and sensitivity kernel analyses. Then, we propose the method to overcome the limitation of the decomposed acoustic wave equation. Finally, we verify our method for a simple syncline model and a modified version of the Marmousi-II model.

## 2. Multiparameter acoustic FWI for VTI media

After we introduce the acoustic modeling and inverse problems based on the decomposed equation for VTI media (Xu et al., 2015), we derive the gradient by the conventional method.

### 2.1.1. Acoustic modeling for VTI media

Among various forward modeling algorithms using the acoustic-approximation for describing anisotropic properties, we choose the elliptic decomposition algorithm suggested by Xu et al. (2015) because of its computational efficiency. In addition, this decoupled equation is easy to apply to complex anisotropic systems such as orthorhombic (Masmoudi and Alkhalifah, 2017) and TTI media (Xu and Zhou, 2014).

In the elliptic decomposition technique, the original pseudo-differential equation (Alkhalifah, 2000) is decomposed into the elliptic and scalar operators. Using this decomposition technique, we can derive the acoustic-approximation of elastic wave equations for 2D VTI media from the dispersion relationship (Xu et al., 2015) as follows:

$$\frac{\partial^2 P}{\partial t^2} = \left( v_0^2 (1 + 2\varepsilon) \frac{\partial^2 P}{\partial x^2} + v_0^2 \frac{\partial^2 P}{\partial z^2} \right) S_e + f_P, \quad (1)$$

$$S_e = \frac{1}{2} \left( 1 + \sqrt{1 - \frac{8(\varepsilon - \delta)n_x^2 n_z^2}{((1 + 2\varepsilon)n_x^2 + n_z^2)}} \right), \quad (2)$$

where  $P$  and  $f_P$  are the pressure wavefields and pressure source, respectively; the anisotropic parameters  $\varepsilon$  and  $\delta$  are the Thomsen parameters (Thomsen, 1986);  $v_0$  denotes the vertical P-wave velocity;  $S_e$  is the elliptic scalar operator; and  $n_x$  and  $n_z$  are the unit vectors used to represent the phase direction. Because the scalar operator is not sensitive to the signs of unit vectors, we can replace the unit vectors of phase directions with the wavefield divergence to avoid taking the inverse Fourier transform of the unit vectors with respect to the space (Xu and Zhou, 2014).

Compared to the coupled systems (Zhou et al., 2006; Duveneck et al., 2008), the decomposed equation does not contain S-wave components and enhances computational efficiency because only one differential equation is solved (Xu and Zhou, 2014). However, although the kinematic information is well preserved compared to the elastic wave equations, the decomposed equation has some amplitude errors when strong anisotropy exists. For the imaging process, the kinematic information is essential to locate subsurface reflectors, whereas for the FWI process, the dynamic information is also crucial to properly estimate subsurface model parameters. Another limitation of the decomposed equation in FWI is that its scattering potentials for anisotropic parameters are quite different from those of the elastic wave equations because the source and receiver wavefields represented by pressure-only are cross-correlated with each other. As a result, the incorrect radiation patterns of the decomposed equation for the anisotropic parameters will influence inversion results.

### 2.1.2. Inverse problem

FWI estimates model parameters from an initial model by minimizing the objective function that measures the difference between observed ( $d^{\text{observed}}$ ) and modeled ( $d^{\text{modeled}}$ ) data over time ( $t$ ). The objective function with the  $l_2$ -norm can be expressed as follows:

$$E = \frac{1}{2} \sum_s \sum_r \int_0^T \left\| d_{p,s}^{\text{modeled}}(r, t, \mathbf{m}) - d_{p,s}^{\text{observed}}(r, t) \right\|^2 dt, \quad (3)$$

where  $s$  and  $r$  are the source and receiver positions, respectively; and  $T$  and the subscript  $P$  denote the total recording time and the pressure component, respectively. The solution, which minimizes the objective function (so-called gradient direction), can be derived by taking a partial derivative of Eq. (3) with respect to the model parameter,  $\mathbf{m}$ . In

multiparameter acoustic VTI FWI for marine streamer data, the modeled data ( $d^{\text{modeled}}$ ) are the pressure fields ( $P$ ) computed for an initial or updated model parameter,  $\mathbf{m}$ , which includes the vertical P-wave velocity ( $v_0$ ) and Thomsen parameters,  $\varepsilon$  and  $\delta$ , as described in Eq. (1).

The gradient direction ( $g_{i,j}$ ) at the  $j^{\text{th}}$  subsurface nodal point for the model parameter ( $i$ : parameter class indicating vertical P-wave velocity or Thomsen parameters) is calculated by the zero-lag cross-correlation between the partial derivative wavefields (so-called Jacobian:  $J_{p,s,j}^i$ ) and the residual wavefields,  $d^{\text{modeled}} - d^{\text{observed}}$ , as follows:

$$g_{i,j} = \frac{\partial E}{\partial m_{i,j}} = \frac{1}{2} \sum_s \sum_r \int_0^T J_{p,s,j}^i(r, t, \mathbf{m}) \cdot [d_{p,s}^{\text{modeled}}(r, t, \mathbf{m}) - d_{p,s}^{\text{observed}}(r, t)] dt, \quad (4)$$

The partial derivative wavefields are obtained by performing  $i \times j$  additional forward modeling at each iteration, which incurs large computational costs. To reduce the computational efforts required for the calculation of the partial derivative wavefields, the adjoint-state method (the so-called backpropagation algorithm) is proposed (Lailly, 1983; Pratt et al., 1998; Plessix, 2006). Using the adjoint-state method, the gradient direction in Eq. (4) can be rewritten as

$$g_{i,j} = \frac{\partial E}{\partial m_{i,j}} = \frac{1}{2} \sum_s \int_0^T f_{p,s,j}^{\text{vir},i}(t, \mathbf{m}) \cdot b_{p,s}^{\text{back}}(T-t, \mathbf{m}) dt, \quad (5)$$

where  $f_{p,s,j}^{\text{vir},i}$  indicates the virtual source of the pressure field at the  $j^{\text{th}}$  subsurface nodal point due to the  $i^{\text{th}}$  model parameter perturbation; and  $b_{p,s}^{\text{back}}$  denotes the backpropagated pressure wavefields, which propagate backward from all the receiver positions using residuals as a source.

To simply derive the virtual source, we transform Eqs. (1) and (2) into the frequency-wavenumber domain:

$$\tilde{s}(\tilde{\mathbf{m}}, \omega, k_x, k_z) \cdot \tilde{p}(\omega, k_x, k_z) = \tilde{f}_p(\omega, k_x, k_z), \quad (6)$$

$$\tilde{s}(\tilde{\mathbf{m}}, \omega, k_x, k_z) = \left( -\omega^2 + \tilde{v}_0^2 \left( (1 + 2\tilde{\varepsilon})k_x^2 + k_z^2 \right) \tilde{S}_e \right), \quad (7)$$

where the tilde indicates the frequency-wavenumber wavefields;  $\tilde{s}$  represents the modeling operator;  $\omega$  is the angular frequency; and  $k_x$  and  $k_z$  are the wavenumbers along the horizontal and vertical axes, respectively. The virtual source can be calculated by taking a partial derivative

of Eq. (6) with respect to each model parameter as follows:

$$\tilde{s}(\tilde{\mathbf{m}}, \omega, k_x, k_z) \frac{\partial \tilde{p}(\omega, k_x, k_z)}{\partial \tilde{m}_{k_x, k_z}^i} = - \left( \frac{\partial \tilde{s}(\tilde{\mathbf{m}}, \omega, k_x, k_z)}{\partial \tilde{m}_{k_x, k_z}^i} \tilde{p}(\omega, k_x, k_z) \right), \quad (8)$$

$$= \tilde{f}_{p,k_x, k_z}^{\text{vir},i}(\omega, k_x, k_z)$$

The term on the right-hand side is the virtual source for the parameter class ( $i$ ), which acts as a source for the partial derivative wavefields ( $\partial p / \partial m$ ). The type of virtual source described by a moment tensor (Oh and Alkhalifah, 2016b) plays an important role in determining the radiation pattern of the partial derivative wavefields due to the model parameter perturbation, which significantly affects the FWI sensitivity to recover the physical properties.

Fig. 1 shows the theoretical PP scattering patterns of the partial derivative wavefields (Gholami et al., 2013a) in terms of the scattering angle at 5 different incident angles. The PP scattering pattern due to the perturbation of P-wave velocity is isotropic over the whole range of incident angles (Fig. 1a). Therefore, the estimation of the P-wave velocity is affected by both low and high wavenumber components, which provides a wide inversion coverage. In Fig. 1b, the scattered wavefields due to the perturbation of anisotropic parameter  $\varepsilon$  show the characteristic of a sine function that mainly propagates horizontally regardless of incident angle. The amplitude of the scattered wavefields is strongest when the incident angle is  $90^\circ$ . Fig. 1c shows that the scattering pattern of perturbation of  $\delta$  depends on the incident angle. The scattered wavefields radiate vertically at large incident angles, while they radiate horizontally at small incident angles. At an incident angle of  $45^\circ$ , the scattered wavefields have an isotropic pattern.

### 2.1.3. Mono-component virtual sources from the decomposed equation

From Eq. (8), the virtual source for the vertical P-wave velocity can be expressed by

$$\tilde{f}_{p,k_x, k_z}^{\text{vir},v_0}(\omega, k_x, k_z) = 2\tilde{v}_0 \left( (1 + 2\tilde{\varepsilon})k_x^2 + k_z^2 \right) \tilde{p}, \quad (9)$$

In a similar way, the virtual source of the Thomsen parameters can be derived by calculating the partial derivative of modeling operator as follows:

$$\tilde{f}_{p,k_x, k_z}^{\text{vir},\varepsilon}(\omega, k_x, k_z) = 2\tilde{v}_0^2 \left( k_x^2 \tilde{S}_e + (k_z^2 + (1 + 2\tilde{\varepsilon})k_x^2) \frac{\partial \tilde{S}_e}{\partial \varepsilon} \right) \tilde{p}, \quad (10)$$

$$\frac{\partial \tilde{S}_e}{\partial \varepsilon} = \frac{-2 \left( (1 - 2\tilde{\varepsilon} + 4\tilde{\delta})n_x^2 + n_z^2 \right) n_x^2 n_z^2}{((1 + 2\tilde{\varepsilon})n_x^2 + n_z^2)^3 (2\tilde{S}_e - 1)}, \quad (11)$$

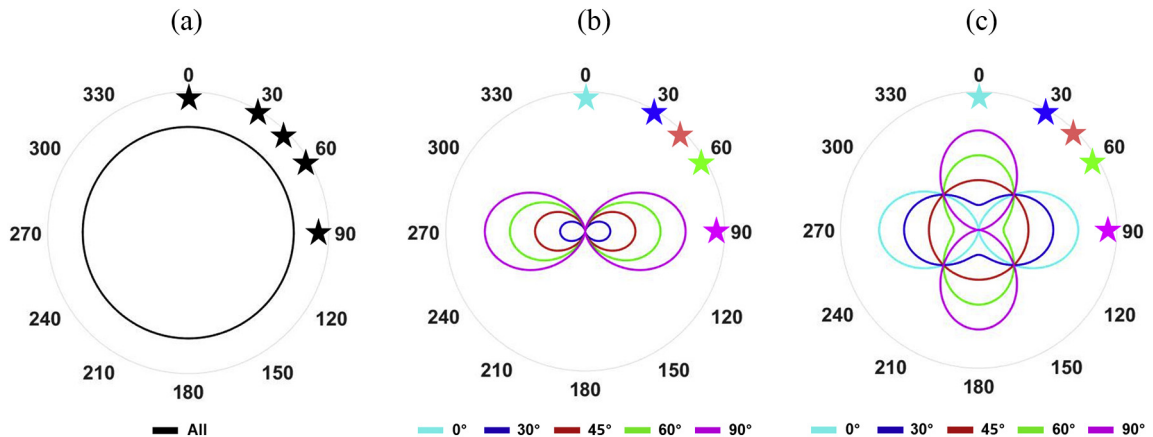
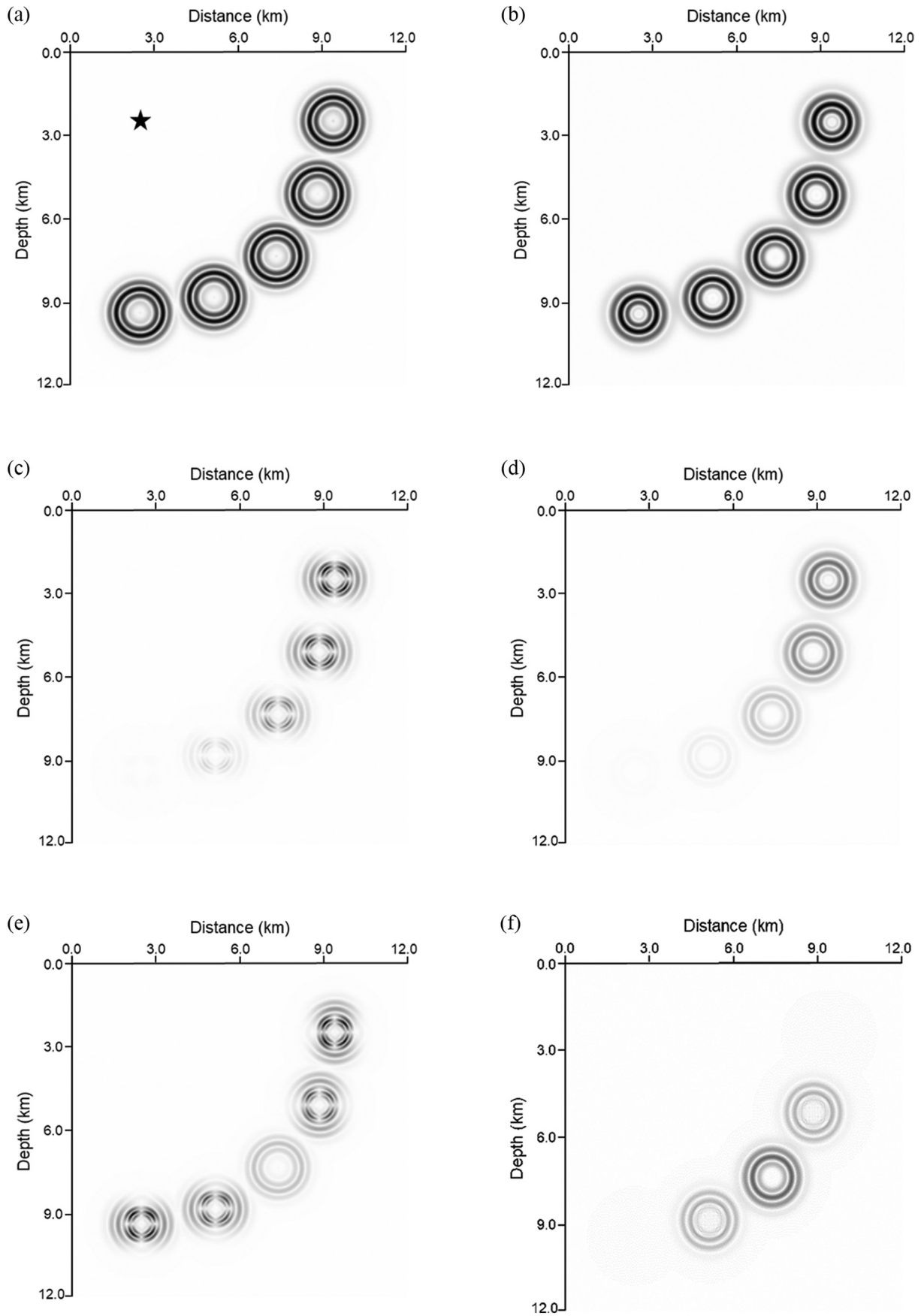


Fig. 1. Theoretical PP scattering patterns for (a) P-wave velocity, (b)  $\varepsilon$  and (c)  $\delta$  at incident angles of  $0^\circ$ ,  $30^\circ$ ,  $45^\circ$ ,  $60^\circ$  and  $90^\circ$ . The different colored stars represent the source locations at each incident angle.



**Fig. 2.** PP scattering patterns of the partial derivative wavefields with the elastic scattering theory for VTI media (a, c and e) and the decomposed equation for acoustic VTI FWI algorithms (b, d and f) for the five pressure virtual sources of the P-wave velocity (a and b),  $\varepsilon$  (c and d) and  $\delta$  (e and f) at different incident angles. The black star indicates the true seismic source location. The partial derivative wavefields from elastic FWI include both PP and PS wavefields, while those of acoustic FWI include only PP wavefields.

$$\tilde{f}_{p,k'_x,k'_z}^{vir,\delta}(\omega, k_x, k_z) = \tilde{v}_0^2 (k_z^2 + (1 + 2\tilde{\epsilon})k_x^2) \frac{\partial \tilde{S}_e}{\partial \delta} \tilde{p}, \quad (12)$$

$$\frac{\partial \tilde{S}_e}{\partial \delta} = \frac{2n_x^2 n_z^2}{((1 + 2\tilde{\epsilon})n_x^2 + n_z^2)^2 (2\tilde{S}_e - 1)}, \quad (13)$$

The virtual sources with respect to the model parameters in the frequency-wavenumber domain can be transformed into the time-space domain by substituting the wavefield divergence for the unit vector of phase direction because the components of virtual sources are not sensitive to the signs of unit vectors.

In Eqs. (9), (10) and (12), the squared wavenumbers, which originate from the spatial derivatives of the pressure wavefields in the space domain, affect the source directional properties that eventually determine the radiation pattern. Because a single component is used in the decomposition-based FWI algorithm, the virtual sources tend to act as an isotropic source.

Fig. 2 shows the scattering patterns of the partial derivative wavefields from the virtual sources at 5 perturbation points in a homogeneous isotropic media. We assume that all the perturbations are located at the points having the same distance from the true seismic source so that all the virtual sources can be simultaneously activated. In Fig. 2, we compare the partial derivative wavefields obtained using the decomposed wave equation (b, d and f) with those obtained by the conventional elastic VTI FWI (a, c and e) including both PP and PS wavefields. In Fig. 2, we observe that the PP radiation patterns from acoustic VTI FWI using the decomposed equation deviate greatly from those of the conventional elastic VTI FWI algorithm. When we apply the decomposed equation, all the parameters similarly show the

isotropic radiation patterns, although the amplitude variations with respect to the incident angle are similar to those of the conventional elastic VTI FWI. Considering that the PP scattering patterns from the elastic VTI FWI are in good agreement with the theoretical PP scattering patterns in Fig. 1, we note that the acoustic VTI FWI using the decomposed equation does not properly describe the radiation patterns for anisotropic parameters.

### 3. Gradient calculation using vector wavefields

The acoustic VTI FWI algorithms using the multicomponent coupled equations proposed by Duvenek et al. (2008) and Gholami et al. (2013a) have presented accurate radiation patterns (Gholami et al., 2013a). By reviewing the virtual sources from the coupled elastic wave equations, we propose a method to provide correct radiation patterns even using the decomposed equation. In our method, we use the displacement vector to compute the virtual sources, unlike the conventional method using the mono-component pressure wavefield.

#### 3.1.1. Virtual sources from the coupled acoustic wave equations

The 2D VTI elastic wave equations based on the particle accelerations can be expressed by the stress-strain relation and the equation of motion as follows:

$$\frac{1}{\rho} a_x = \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xz}}{\partial z}, \quad (14)$$

$$\frac{1}{\rho} a_z = \frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{zz}}{\partial z}, \quad (15)$$

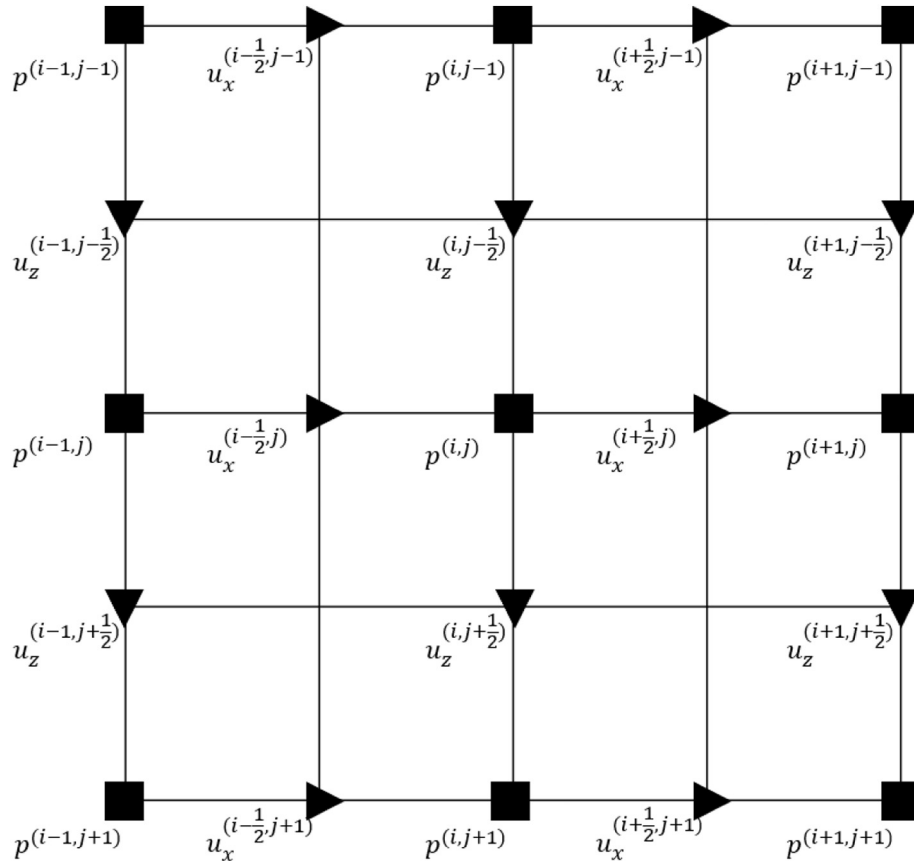
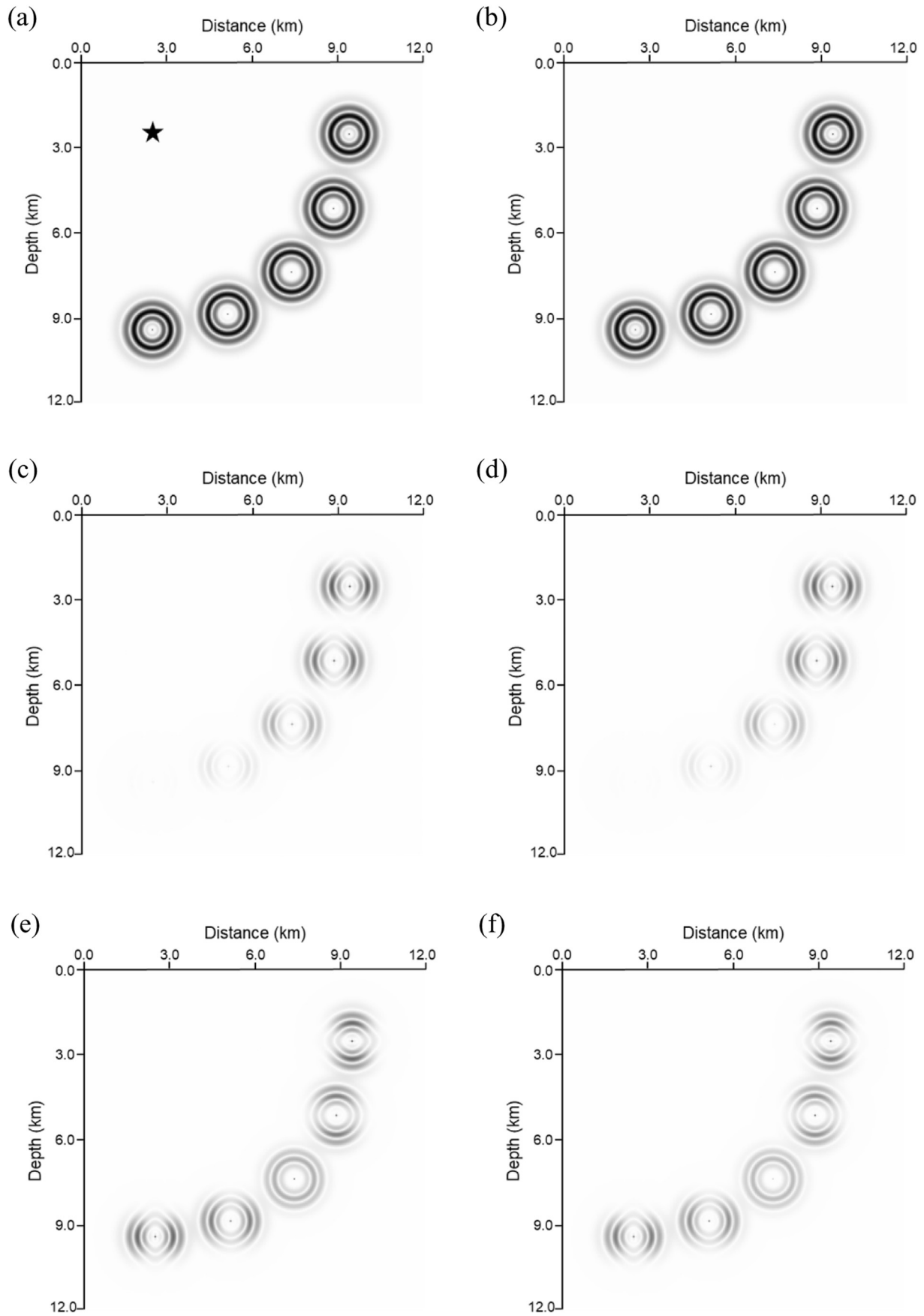
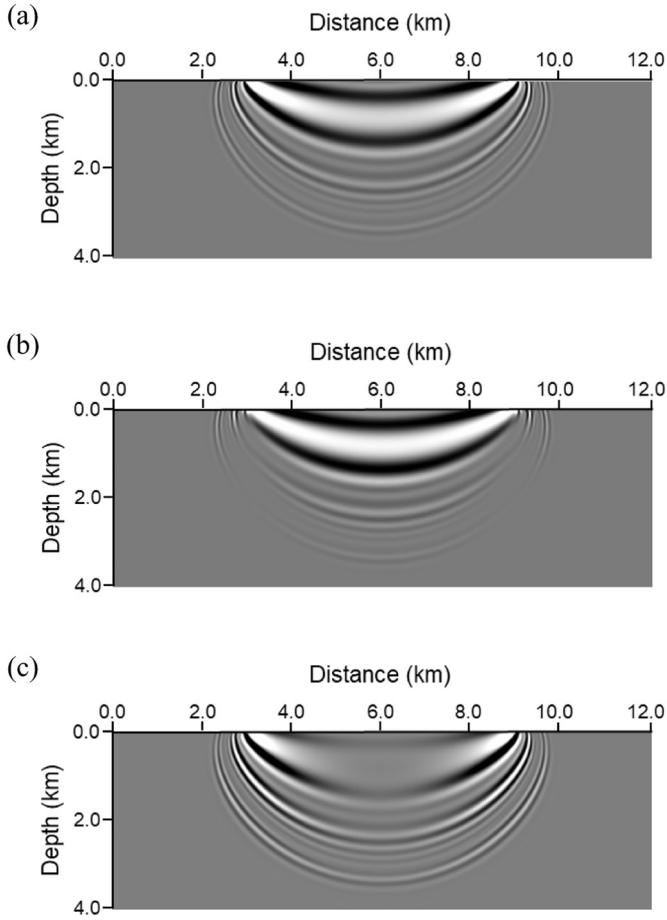


Fig. 3. Staggered grid set to obtain vector wavefields from the pressure wavefields (P). Model parameters (the P-wave velocity and Thomsen parameters) are placed in the same grid as the pressure wavefields.



**Fig. 4.** PP scattering patterns of the partial derivative wavefields with the coupled equation of Gholami et al. (2013a; a, c and e) and the proposed method with displacement wavefields using the decomposed equation (b, d and f) for the five pressure virtual sources of P-wave velocity (a and b),  $\epsilon$  (c and d) and  $\delta$  (e and f) at different incident angles.



**Fig. 5.** Sensitivity kernels using the coupled equation for the perturbations of (a) P-wave velocity, (b)  $\varepsilon$  and (c)  $\delta$ .

$$\frac{\partial^2 \sigma_{xx}}{\partial t^2} = C_{11} \frac{\partial a_x}{\partial x} + C_{13} \frac{\partial a_z}{\partial z} + f_{\sigma_{xx}}, \quad (16)$$

$$\frac{\partial^2 \sigma_{zz}}{\partial t^2} = C_{13} \frac{\partial a_x}{\partial x} + C_{33} \frac{\partial a_z}{\partial z} + f_{\sigma_{zz}}, \quad (17)$$

$$\frac{\partial^2 \sigma_{xz}}{\partial t^2} = C_{55} \left( \frac{\partial a_z}{\partial x} + \frac{\partial a_x}{\partial z} \right), \quad (18)$$

where  $a_x$  and  $a_z$  are the particle accelerations;  $\sigma_{xx}$ ,  $\sigma_{zz}$ , and  $\sigma_{xz}$  are two normal stresses and shear stress, respectively; and  $C_{ij}$  are the stiffness coefficients. In acoustic media with homogeneous density, the 2D VTI elastic wave equations in terms of the stress wavefields can be rewritten as follows (Duveneck et al., 2008):

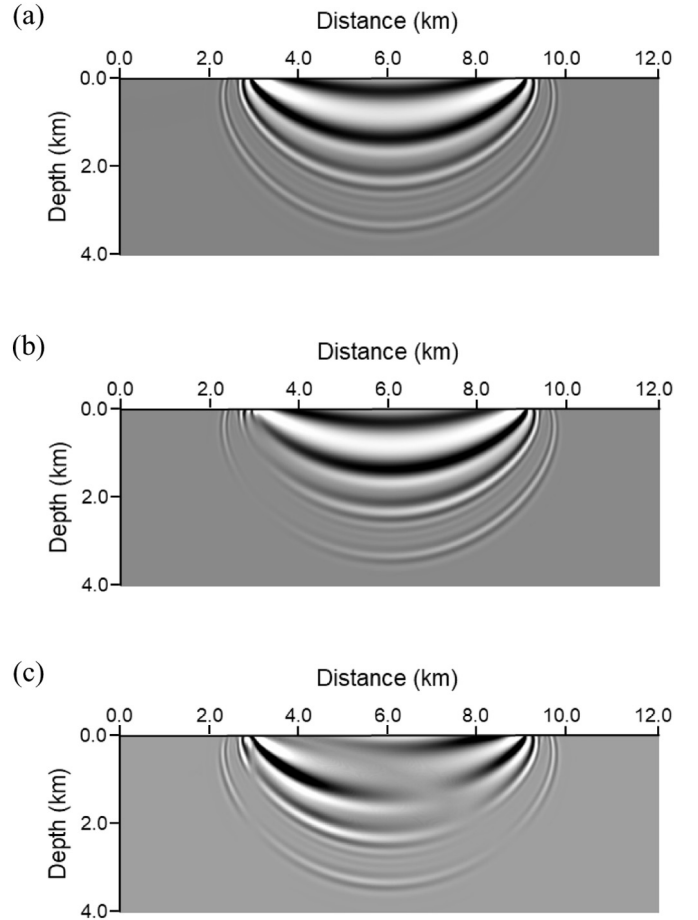
$$\frac{\partial^2 \sigma_{xx}}{\partial t^2} = \left( v_0^2 (1 + 2\varepsilon) \frac{\partial^2 \sigma_{xx}}{\partial x^2} + v_0^2 \sqrt{1 + 2\delta} \frac{\partial^2 \sigma_{zz}}{\partial z^2} \right) + f_{\sigma_{xx}}, \quad (19)$$

$$\frac{\partial^2 \sigma_{zz}}{\partial t^2} = \left( v_0^2 \sqrt{1 + 2\delta} \frac{\partial^2 \sigma_{xx}}{\partial x^2} + v_0^2 \frac{\partial^2 \sigma_{zz}}{\partial z^2} \right) + f_{\sigma_{zz}}, \quad (20)$$

where the normal stresses  $\sigma_{xx}$  and  $\sigma_{zz}$  reduce to the pressure fields ( $P$ ) in the decomposed equation.

In the frequency-wavenumber domain, the stress-based coupled acoustic wave equation is expressed as

$$\tilde{\mathbf{K}}_\sigma(\tilde{\mathbf{m}}, \omega, k_x, k_z) \cdot \tilde{\boldsymbol{\sigma}}(\omega, k_x, k_z) = \tilde{\mathbf{f}}_\sigma(\omega, k_x, k_z), \quad (21)$$



**Fig. 6.** Sensitivity kernels using the decomposed equation for the perturbations of (a) P-wave velocity, (b)  $\varepsilon$  and (c)  $\delta$ .

which can be rewritten in a matrix form as:

$$\begin{pmatrix} -\omega^2 + \tilde{v}_0^2 (1 + 2\varepsilon) k_x^2 & \tilde{v}_0^2 (\sqrt{1 + 2\delta}) k_z^2 \\ \tilde{v}_0^2 (\sqrt{1 + 2\delta}) k_x^2 & -\omega^2 + \tilde{v}_0^2 k_z^2 \end{pmatrix} \begin{pmatrix} \tilde{\sigma}_{xx} \\ \tilde{\sigma}_{zz} \end{pmatrix} = \begin{pmatrix} \tilde{f}_{\sigma_{xx}} \\ \tilde{f}_{\sigma_{zz}} \end{pmatrix}, \quad (22)$$

Eq. (22) shows that the stress-based coupled acoustic wave equation is not self-adjoint; therefore, the gradient cannot be calculated by the adjoint-state method as in Eq. (5).

Gholami et al. (2013a) suggested the coupled acoustic wave equations based on the vector wavefield (displacement), which are expressed by:

$$\frac{\partial^2 u_x}{\partial t^2} = \frac{\partial}{\partial x} \left( v_0^2 (1 + 2\varepsilon) \frac{\partial u_x}{\partial x} + v_0^2 \sqrt{1 + 2\delta} \frac{\partial u_z}{\partial z} \right) + \frac{\partial f_{\sigma_{xx}}}{\partial x}, \quad (23)$$

$$\frac{\partial^2 u_z}{\partial t^2} = \frac{\partial}{\partial z} \left( v_0^2 \sqrt{1 + 2\delta} \frac{\partial u_x}{\partial x} + v_0^2 \frac{\partial u_z}{\partial z} \right) + \frac{\partial f_{\sigma_{zz}}}{\partial z}, \quad (24)$$

where  $u_x$  and  $u_z$  are the horizontal and vertical particle displacements, respectively. This displacement-based coupled acoustic wave equation is self-adjoint; therefore, we can obtain the gradient direction using the adjoint-state method with much lower computational costs. In the frequency-wavenumber domain, the displacement-based coupled acoustic wave equation is expressed as

$$\tilde{\mathbf{K}}_u(\tilde{\mathbf{m}}, \omega, k_x, k_z) \cdot \tilde{\mathbf{u}}(\omega, k_x, k_z) = \tilde{\mathbf{f}}_u(\omega, k_x, k_z), \quad (25)$$

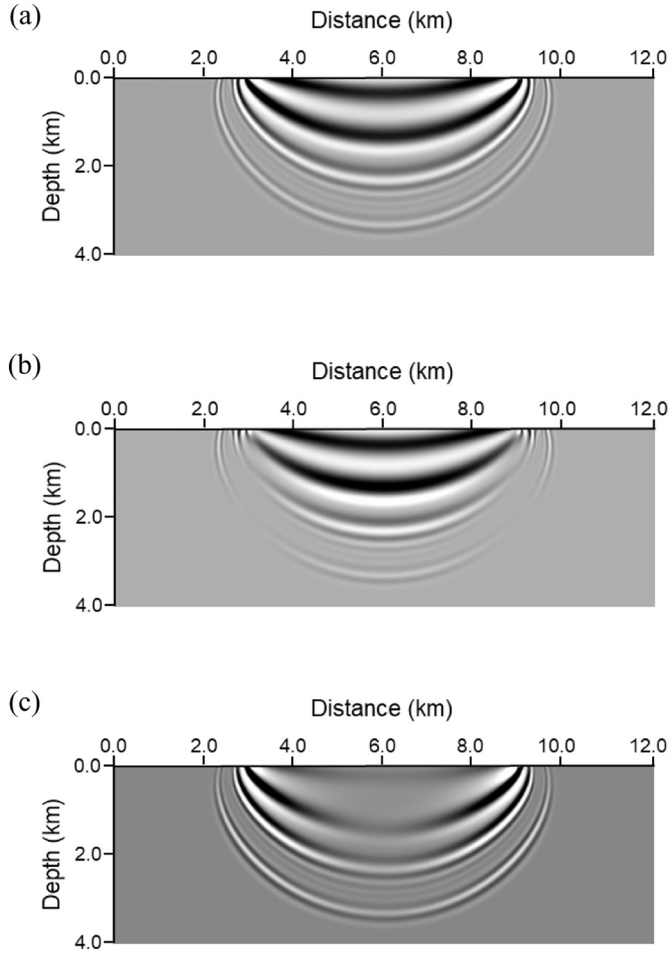


Fig. 7. Sensitivity kernels using the proposed method for the perturbations of (a) P-wave velocity, (b)  $\varepsilon$  and (c)  $\delta$ .

which can be rewritten in a matrix form:

$$\begin{pmatrix} -\omega^2 - \tilde{v}_0^2(1 + 2\tilde{\varepsilon})k_x^2 & -\tilde{v}_0^2(\sqrt{1 + 2\tilde{\delta}})k_x k_z \\ -\tilde{v}_0^2(\sqrt{1 + 2\tilde{\delta}})k_z k_x & -\omega^2 - \tilde{v}_0^2 k_z^2 \end{pmatrix} \begin{pmatrix} \tilde{u}_x \\ \tilde{u}_z \end{pmatrix} = \begin{pmatrix} ik_x \tilde{f}_{\sigma_{xx}} \\ ik_z \tilde{f}_{\sigma_{zz}} \end{pmatrix}, \quad (26)$$

In the same way described in the previous section, the displacement components of the virtual sources for the P-wave velocity,  $\varepsilon$  and  $\delta$  can be obtained by

$$\begin{pmatrix} \tilde{f}_{u_x, k'_x, k'_z}^{\text{vir}, v_0} \\ \tilde{f}_{u_z, k'_x, k'_z}^{\text{vir}, v_0} \end{pmatrix} = \begin{pmatrix} 2\tilde{v}_0(1 + 2\tilde{\varepsilon})k_x^2 & 2\tilde{v}_0(\sqrt{1 + 2\tilde{\delta}})k_x k_z \\ 2\tilde{v}_0(\sqrt{1 + 2\tilde{\delta}})k_z k_x & 2\tilde{v}_0 k_z^2 \end{pmatrix} \begin{pmatrix} \tilde{u}_x \\ \tilde{u}_z \end{pmatrix}, \quad (27)$$

$$\begin{pmatrix} \tilde{f}_{u_x, k'_x, k'_z}^{\text{vir}, \varepsilon} \\ \tilde{f}_{u_z, k'_x, k'_z}^{\text{vir}, \varepsilon} \end{pmatrix} = \begin{pmatrix} 2\tilde{v}_0^2 k_x^2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{u}_x \\ \tilde{u}_z \end{pmatrix}, \quad (28)$$

$$\begin{pmatrix} \tilde{f}_{u_x, k'_x, k'_z}^{\text{vir}, \delta} \\ \tilde{f}_{u_z, k'_x, k'_z}^{\text{vir}, \delta} \end{pmatrix} = \begin{pmatrix} 0 & \frac{\tilde{v}_0^2}{\sqrt{1 + 2\tilde{\delta}}} k_x k_z \\ \frac{\tilde{v}_0^2}{\sqrt{1 + 2\tilde{\delta}}} k_z k_x & 0 \end{pmatrix} \begin{pmatrix} \tilde{u}_x \\ \tilde{u}_z \end{pmatrix}, \quad (29)$$

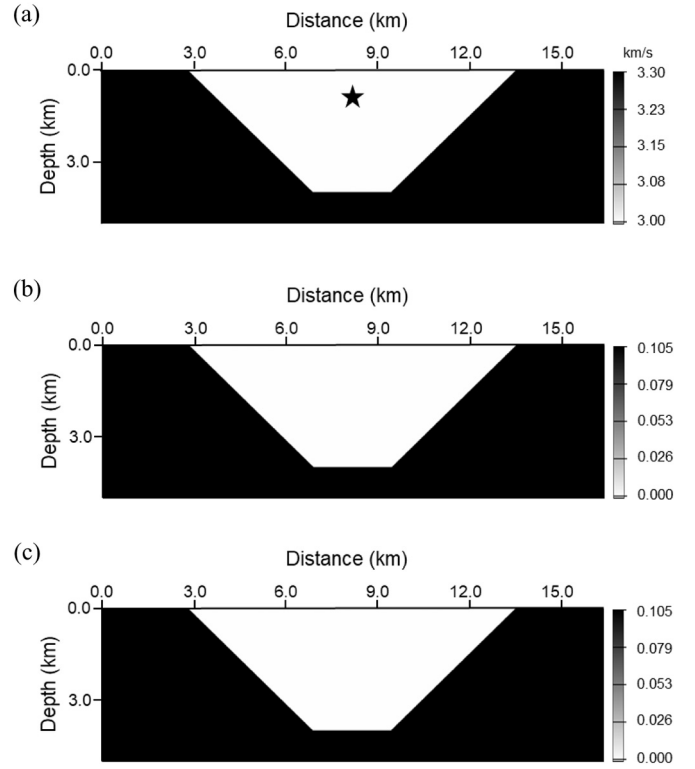


Fig. 8. The simple syncline model: (a) P-wave velocity, (b)  $\varepsilon$  and (c)  $\delta$ . The black star indicates the seismic source location, and receivers are placed at surface.

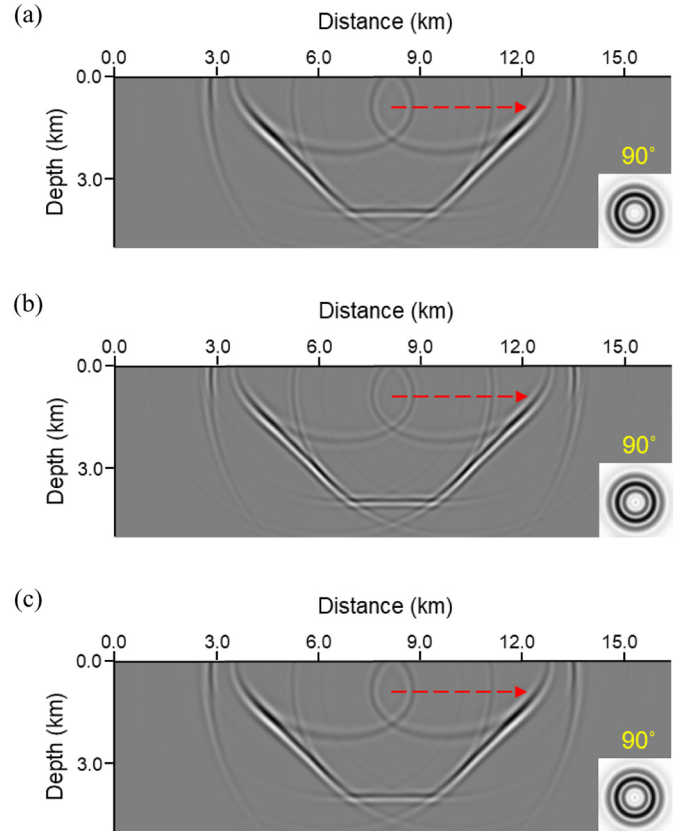
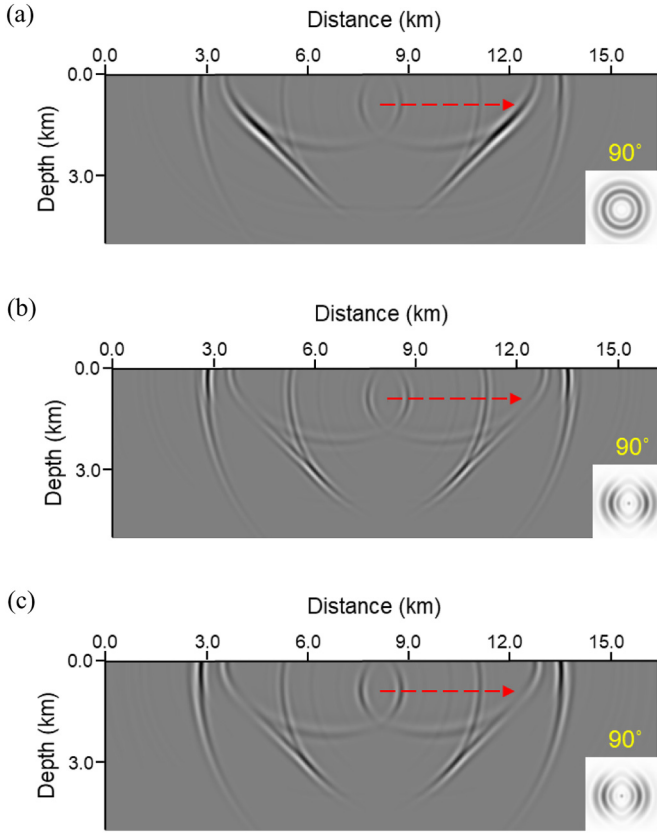


Fig. 9. Gradient directions of P-wave velocity for the simple syncline model using (a) the decomposed equation, (b) coupled wave equation and (c) the proposed approach with displacement wavefields. The bottom figure shows the radiation pattern for the P-wave velocity at an incident angle of  $90^\circ$  from Figs. 2 and 4. The red dashed arrows indicate the incident wavefields.



**Fig. 10.** Gradient directions of  $\varepsilon$  for the simple syncline model using (a) the decomposed equation, (b) coupled wave equation and (c) the proposed approach with displacement wavefields. The bottom figure shows the radiation pattern for the P-wave velocity at an incident angle of  $90^\circ$  from Figs. 2 and 4. The red dashed arrows indicate the incident raypaths.

The virtual sources in the time-space domain can be obtained after the double inverse Fourier transform. The multiplies of wavenumber components ( $k_x k_z$ ,  $k_x^2$  and  $k_z^2$ ) in the virtual sources of Eqs. (27), (28) and (29) originate from the spatial derivatives of the wave equation in the space domain and determine the radiation patterns of the partial derivative wavefields describing the moment-tensor sources (Oh and Alkhalifah, 2016b). Using the adjoint-state method, the gradient direction can be obtained by the displacement components of wavefields as follows:

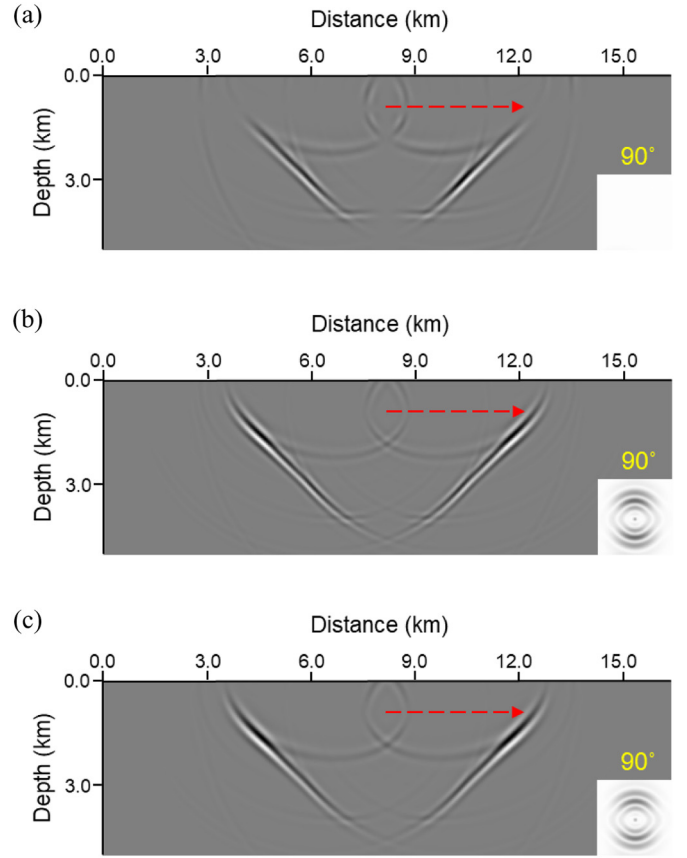
$$g_{i,j} = \frac{1}{2} \sum_s \int_0^T \left\{ f_{u_x, s, j}^{vir, i}(t, \mathbf{m}) \cdot b_{u_x, s}^{back}(T-t, \mathbf{m}) + f_{u_z, s, j}^{vir, i}(t, \mathbf{m}) \cdot b_{u_z, s}^{back}(T-t, \mathbf{m}) \right\} dt, \quad (30)$$

where  $b_{u, s}^{back}$  is the particle displacement component of the backpropagated wavefields obtained by the displacement residuals converted from pressure residuals with the equation of motion in acoustic media (Gholami et al., 2013a).

### 3.1.2. The proposed method based on vector wavefields

To overcome the limitation of the decomposed equation with mono-component, we use vector wavefields for the computation of the gradient direction. The vector wavefields are obtained by converting the mono-component pressure into the displacement vector.

Under the assumption that media are acoustic and isotropic (where  $P = \sigma_{xx} = \sigma_{zz}$ ), the pressure component can be approximately



**Fig. 11.** Gradient directions of  $\delta$  for the simple syncline model using (a) the mono-component decomposed equation, (b) coupled wave equation and (c) the proposed approach with displacement wavefields. The bottom figure shows the radiation pattern for the P-wave velocity at an incident angle of  $90^\circ$  from Figs. 2 and 4. The red dashed arrows indicate the incident raypaths.

converted to the particle displacements using the equation of motion in Eqs. (14) and (15) as follows:

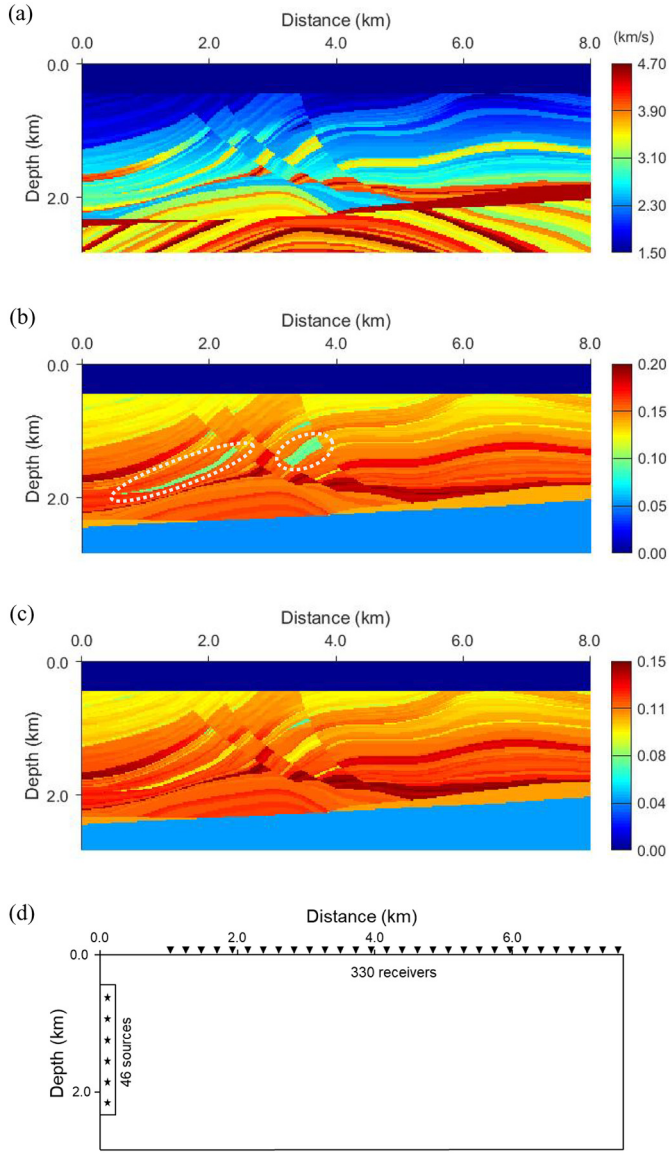
$$\tilde{u}_x^{-P} = -\frac{1}{\omega^2} \tilde{a}_x^{-P} \approx -\frac{1}{\omega^2} \frac{\partial \tilde{P}}{\partial x}, \quad (31)$$

$$\tilde{u}_z^{-P} = -\frac{1}{\omega^2} \tilde{a}_z^{-P} \approx -\frac{1}{\omega^2} \frac{\partial \tilde{P}}{\partial z}, \quad (32)$$

where  $\tilde{u}^{-P}$  and  $\tilde{a}^{-P}$  are the particle displacements and particle accelerations, respectively, converted from the modeled pressure wavefields ( $P$ ) using the decomposed equation. The vector components are obtained at the staggered grids, which are placed at the midpoints between the pressure wavefields (see Fig. 3). With the converted particle displacements, the new vector-based virtual sources can be obtained by

$$\begin{pmatrix} \tilde{f}_{u_x^{-P}, k'_x, k'_z}^{vir, v_0} \\ \tilde{f}_{u_z^{-P}, k'_x, k'_z}^{vir, v_0} \end{pmatrix} = \begin{pmatrix} 2\tilde{v}_0(1+2\tilde{\varepsilon})k_x^2 & 2\tilde{v}_0(\sqrt{1+2\tilde{\varepsilon}})k_x k_z \\ 2\tilde{v}_0(\sqrt{1+2\tilde{\varepsilon}})k_z k_x & 2\tilde{v}_0 k_z^2 \end{pmatrix} \begin{pmatrix} \tilde{u}_x^{-P} \\ \tilde{u}_z^{-P} \end{pmatrix}, \quad (33)$$

$$\begin{pmatrix} \tilde{f}_{u_x^{-P}, k'_x, k'_z}^{vir, \varepsilon} \\ \tilde{f}_{u_z^{-P}, k'_x, k'_z}^{vir, \varepsilon} \end{pmatrix} = \begin{pmatrix} 2\tilde{v}_0^2 k_x^2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{u}_x^{-P} \\ \tilde{u}_z^{-P} \end{pmatrix}, \quad (34)$$



**Fig. 12.** The modified Marmousi-II model: (a) P-wave velocity, (b)  $\epsilon$ , (c)  $\delta$ , and (d) the locations of sources and receivers, which are designed for a borehole experiment.

$$\begin{pmatrix} \tilde{f}_{u_x^p, k'_x, k'_z}^{vir, \delta} \\ \tilde{f}_{u_z^p, k'_x, k'_z}^{vir, \delta} \end{pmatrix} = \begin{pmatrix} 0 & \frac{\tilde{v}_0^2}{\sqrt{1+2\delta}} k_x k_z \\ \frac{\tilde{v}_0^2}{\sqrt{1+2\delta}} k_z k_x & 0 \end{pmatrix} \begin{pmatrix} \tilde{u}_x^{p-} \\ \tilde{u}_z^{p-} \end{pmatrix}, \quad (35)$$

In the inversion process, we first simulate the forward and backpropagated pressure wavefields using the decomposed equation and then convert them to the particle displacement components using Eqs. (31) and (32). The adjoint form with the modified displacement virtual sources can be expressed by

$$\tilde{f}^i = (\tilde{s}^{-1}) \mathbf{B} \tilde{\mathbf{f}}_u^{vir, i}, \quad (36)$$

$$\mathbf{B} = \left( \tilde{v}_0^2 (1 + 2\epsilon) k_x \tilde{v}_0^2 k_z \right) \tilde{\mathbf{S}}_e, \quad (37)$$

$$\tilde{\mathbf{g}}_i = \left( \tilde{\mathbf{f}}_{u^p}^{vir, i} \right)^T \tilde{\mathbf{b}}_{u^p}^{back}, \quad (38)$$

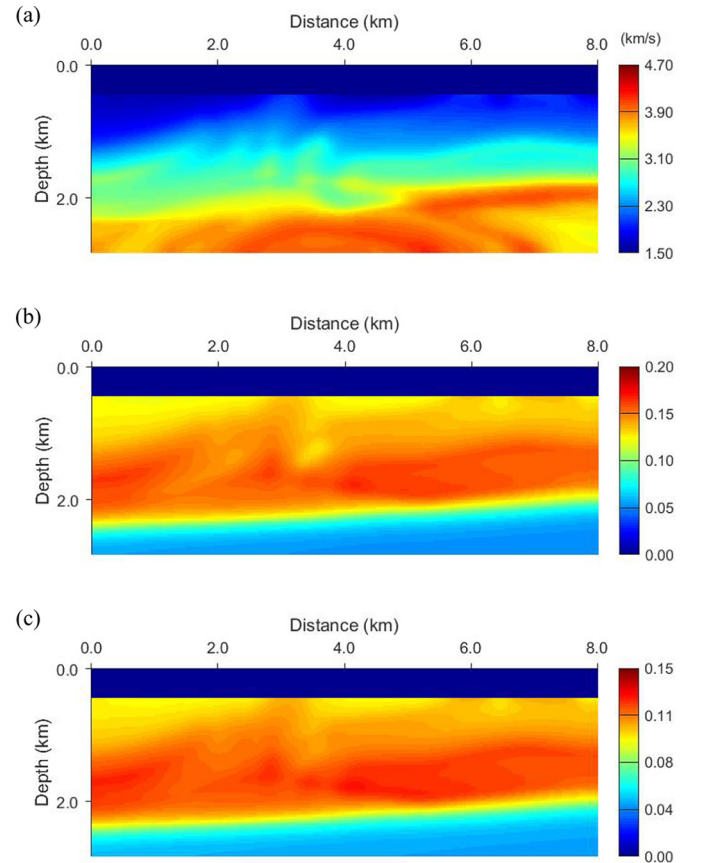
$$\tilde{\mathbf{b}}_{u^p}^{back} = \mathbf{B}^T \tilde{\mathbf{b}}_p^{back}, \quad (39)$$

The operator  $\mathbf{B}$  represents the conversion matrix from the particle displacements to pressure components. To apply the adjoint-state method, the transpose of  $\mathbf{B}$  is required to convert the backpropagated pressure wavefields to displacement components. Finally, we can obtain the gradient direction using the particle displacement components as in Eq. (30) as follows:

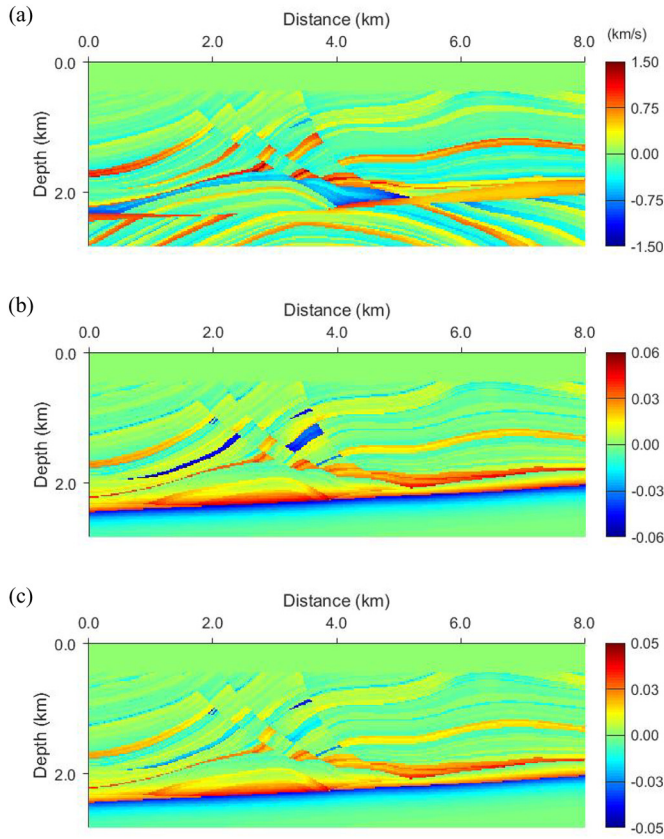
$$\mathbf{g}_{i,j} = \frac{1}{2} \sum_s \int_0^T \left\{ \tilde{f}_{u_x^p, s, j}^{vir, i}(t, \mathbf{m}) \cdot \tilde{b}_{u_x^p, s}^{back}(T-t, \mathbf{m}) + \tilde{f}_{u_z^p, s, j}^{vir, i}(t, \mathbf{m}) \cdot \tilde{b}_{u_z^p, s}^{back}(T-t, \mathbf{m}) \right\} dt, \quad (40)$$

The converted vector wavefields of the virtual sources and backpropagated wavefields can be the particle velocity or acceleration with a proper conversion operator such as a pressure-to-velocity conversion operator or a pressure-to-acceleration conversion operator as a substitute for the particle displacement.

Fig. 4 shows the PP scattering patterns of the partial derivative wavefields for different incident angles obtained with the FWI algorithms with coupled wave equation and the proposed approach. In Fig. 4, we see that both methods provide PP radiation patterns comparable to those from elastic VTI FWI in Fig. 2. Figs. 5, 6 and 7 show the PP sensitivity kernels from the three different acoustic VTI FWI algorithms. For P-wave velocity perturbation (Figs. 5a, 6a and 7a), all the methods provide accurate sensitivity kernels that have clear banana-shaped diving wave kernel and migration isochrone. However, the sensitivity kernels of the anisotropic parameters (Fig. 6)



**Fig. 13.** Initial models used for FWI of the modified Marmousi-II model: (a) P-wave velocity, (b)  $\epsilon$  and (c)  $\delta$ .



**Fig. 14.** Differences between the true and initial models for the modified Marmousi-II model: (a) P-wave velocity, (b)  $\epsilon$  and (c)  $\delta$ .

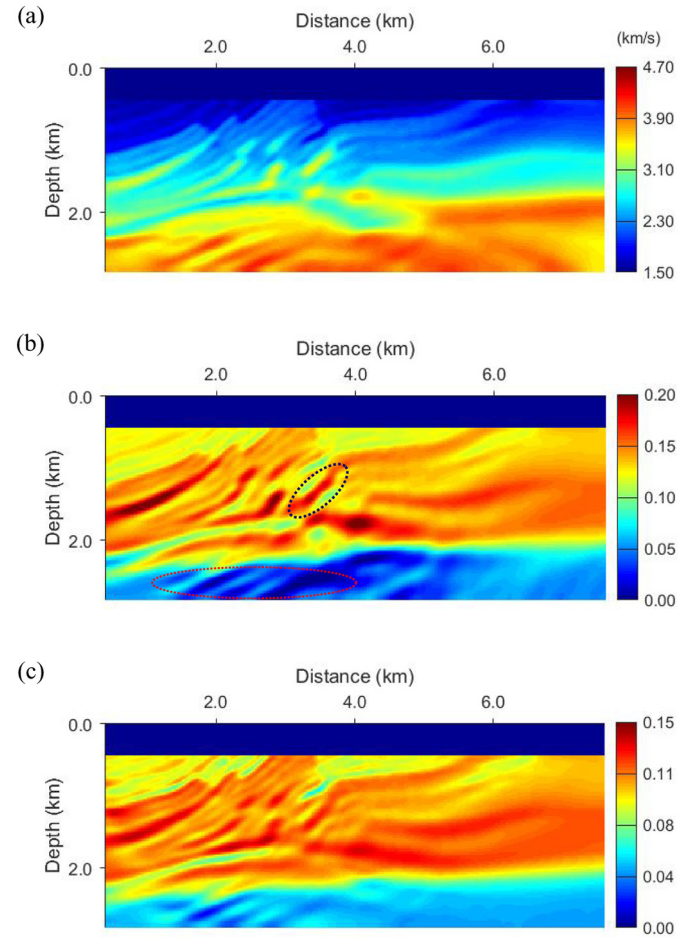
using the decomposed wave equation have asymmetric kernels compared to the other methods. This is due to the incorrect isotropic scattering patterns of the partial derivative wavefields (Fig. 2d and f), which makes the receiver-side kernel resemble the sensitivity kernel of the P-wave velocity. The proposed approach provides almost the same sensitivity kernels as those from the coupled wave equation with lower computational costs.

As aforementioned, the decomposed wave equation provides the computational efficiency, does not suffer from the shear-wave artifacts and has a good extension to tilted anisotropic media, although the radiation patterns of the partial derivative wavefields are not correctly simulated. On the other hand, the coupled wave equation provides more accurate wave propagation but suffers higher computational costs and the shear-wave artifacts. Using the proposed approach, the forward modeling is performed by the decomposed wave equation, and inverse modeling is performed based on the vector wavefields to accurately describe the radiation pattern of the partial derivative wavefields. In the whole FWI procedure, solving the forward modeling is the most time-consuming part, while calculation of the gradient requires less computing time. For this reason, the proposed approach has all the advantages of the conventional anisotropic FWI algorithms.

**Table 1**

Inversion settings used for numerical tests of the modified Marmousi-II model.

| Model size              | No. of shots | No. of receivers | Shot interval | Recording time | Peak frequency | Type of source     |
|-------------------------|--------------|------------------|---------------|----------------|----------------|--------------------|
| 7.6 km $\times$ 2.82 km | 46           | 330              | 40 m          | 4 s            | 5 Hz           | The Ricker wavelet |



**Fig. 15.** FWI results obtained by the mono-component virtual source: (a) P-wave velocity, (b)  $\epsilon$  and (c)  $\delta$ .

#### 4. Numerical examples

We compare the FWI performances of the conventional and proposed approaches based on the decomposed equation for two numerical examples; one is a simple syncline model with surface seismic acquisition, and the other is a modified version of the Marmousi-II model with borehole seismic acquisition.

##### 4.1.1. The simple syncline model

We examine how the different radiation patterns affect the gradient direction for the syncline synthetic model shown in Fig. 8. A single source is applied at a position of (8.5 km, 1 km), and receivers are located at all nodal points of the surface. A Ricker wavelet with a peak frequency of 5 Hz is used as a source wavelet. For initial guesses, we use homogeneous models assuming that the P-wave velocity is 3 km/s and the anisotropic parameters are zero. In Figs. 9, 10 and 11, we compare the gradients obtained by the coupled equation (vector virtual sources) and the conventional (mono-component virtual sources) and new (vector virtual sources) decomposition-based

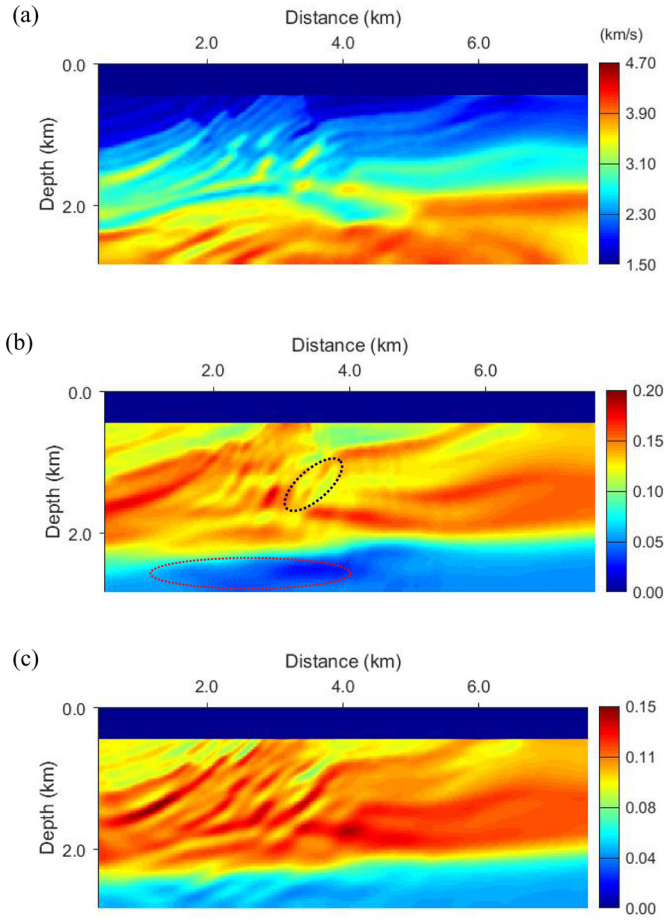


Fig. 16. FWI results obtained by the proposed method: (a) P-wave velocity, (b)  $\epsilon$  and (c)  $\delta$ .

methods with each other. For P-wave velocity, the gradients obtained by the three methods are similar to each other, because all the three methods produce isotropic radiation patterns (see Fig. 9). Fig. 10 shows the gradients of the anisotropic parameter  $\epsilon$ . Recall that the scattered wavefields of  $\epsilon$  propagate horizontally regardless of incident angle according to the theoretical PP radiation pattern (Fig. 1). This indicates that it can be difficult to detect structures in the gradient directions when the incident angle is  $90^\circ$  and receivers are located above the source position. In Fig. 10a, the gradient of  $\epsilon$  obtained using the mono-component virtual sources clearly shows the inclined structure at an incident angle of  $90^\circ$  because its radiation pattern is isotropic, which causes the trade-off between P-wave velocity and  $\epsilon$ . However, in Fig. 10b and c, the gradients of  $\epsilon$  using the vector wavefields have weak signals at an incident angle of  $90^\circ$  (see the structure denoted by the red arrows), which is consistent with the theoretical elastic radiation pattern. As the scattered wavefields of  $\epsilon$  generated with narrow incident angles have little energy, the gradients of  $\epsilon$  obtained using all three methods cannot properly image the horizontal reflector of the syncline structure. Fig. 11 shows the gradient of  $\delta$ . Remember that for an incident angle of  $90^\circ$ , the scattered wavefields of  $\delta$  using the vector-based methods propagate vertically, whereas those obtained using the mono-component virtual source have little energy. In the mono-component case shown in Fig. 11a, the inclined structures are not detected because of insufficient data at wide incident angles. With the vector-based virtual sources, the vertically spreading wavefields can be recorded at the surface receivers, in contrast to the result of  $\epsilon$ ; therefore, the inclined structures

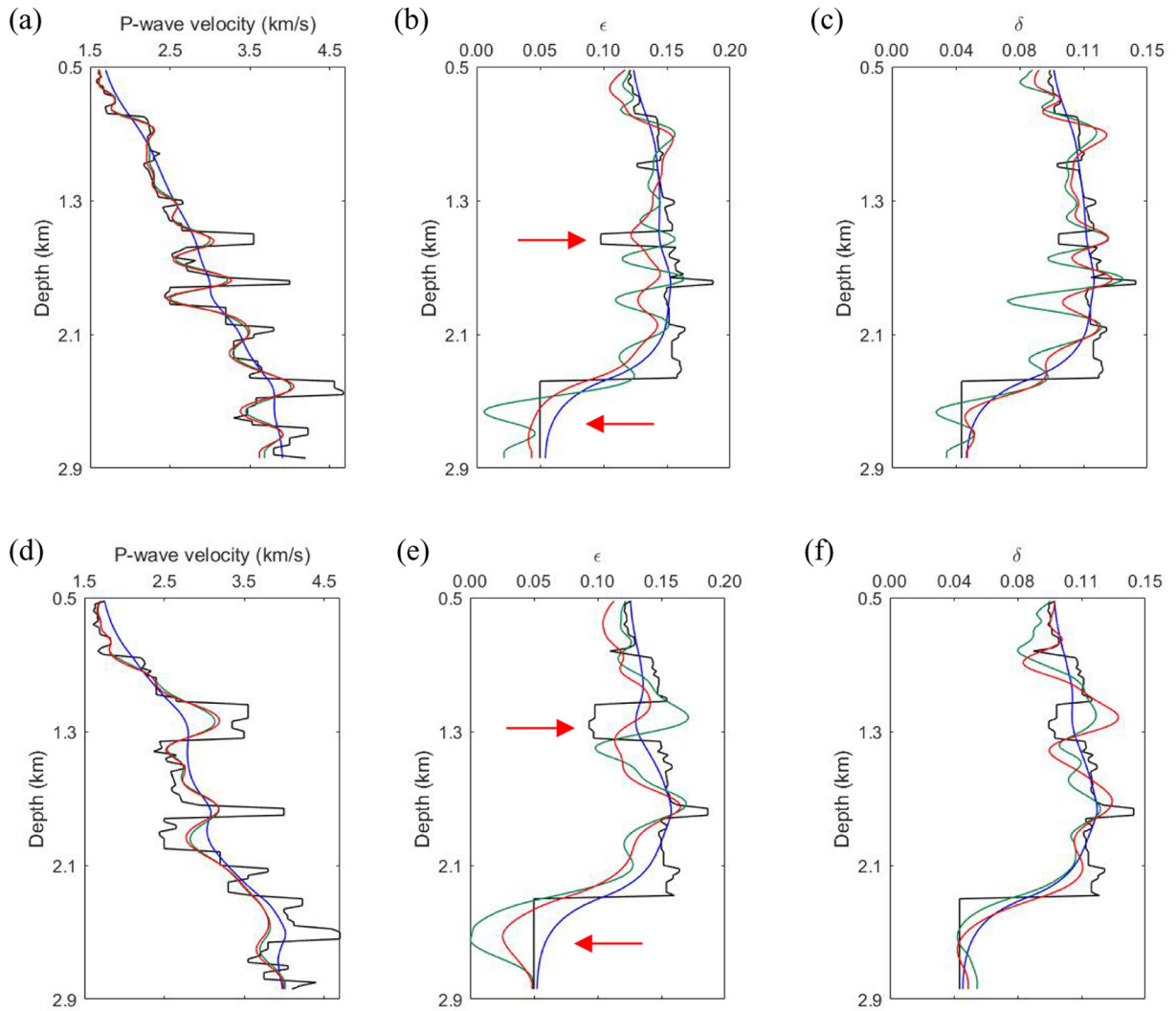
are described well in the gradient of  $\delta$  (see the red arrows in Fig. 11b and c).

#### 4.1.2. The modified Marmousi-II model: borehole experiment

We apply FWI to a modified version of the Marmousi-II model (Fig. 12) to demonstrate different behaviors of the anisotropic parameter perturbations in complicated media. To examine the trade-offs between the P-wave velocity and anisotropic parameters, we assume the two anomalous layers (represented by the white dashed ovals in Fig. 12b), where the anisotropic parameters have a reverse tendency to that of P-wave velocity. We also set the homogeneous anisotropic parameter at the bottom part of the model for both  $\epsilon$  and  $\delta$ , from which we can investigate the trade-off between the three parameters. To clearly show the difference between the scattered wavefields obtained using the conventional and proposed methods, we assume the borehole experiment.

The CPML (convolutional perfectly matched layer) is applied to remove the unwanted reflections from the artificial boundaries (Komatitsch and Martin, 2007; Ping et al., 2014). A Ricker wavelet with a dominant frequency of 5 Hz is employed as a source time function. Sources are applied at depths from 0.46 to 2.3 km with an interval of 40 m in the left-hand side of the model, and the receivers are located at the surface (see Fig. 12d). Initial guesses are smoothed versions of the true models as shown in Fig. 13, which were obtained by the module 'SMOOTH2' in Seismic Unix with the smoothing parameters of 15 in the horizontal and vertical axes. Fig. 14 shows differences between the true and initial models, which correspond to the correct gradient directions for each parameter and indicate the regions to be updated with the gradient direction. Table 1 shows the parameters used for FWI. The diagonal of the pseudo-Hessian matrix (Shin et al., 2001) is used as a preconditioner for the gradient direction. We use constant step lengths for each parameter, which were determined through numerical tests. We invert all the parameters simultaneously using the conventional (using the mono-component virtual sources) and proposed (using the vector virtual sources) methods. In the inversion process, regions around sources are not updated.

Figs. 15 and 16 show inversion results obtained by the mono-component virtual sources and the vector virtual sources, respectively. In Figs. 15a and 16a, we see that the P-wave velocity, which is the primary parameter, is well inverted using both methods because the gradient covers the full-range of incident angles due to their isotropic radiation patterns. However, the structures at the bottom part are not properly inverted because of the inherent coverage limit encountered in the borehole seismic design. In Fig. 15b,  $\epsilon$  recovered using the mono-component virtual source shows a severe trade-off with P-wave velocity, particularly in high-wavenumber parts. We observe that  $\epsilon$  is updated in the wrong direction at the anomalous areas (the black dashed oval at the center in Fig. 15b) and the homogeneous layer at the bottom deforms, mimicking the structure of the P-wave velocity model below the unconformity (the red dashed oval at the bottom in Fig. 15b). In contrast, Fig. 16b shows that  $\epsilon$  is weakly affected by the P-wave velocity but updated along the correct direction by our method. In Fig. 16b, we also observe that  $\epsilon$  is smoothly updated. This is because the scattered wavefields horizontally propagate in our method, which causes some trade-offs with P-wave velocity in low-wavenumber parts recovered from wide opening angle components. On the other hand, according to the elastic radiation pattern of  $\delta$  in Fig. 1c, we expect that the incident wavefields at various angles can correlate with the backpropagated wavefields in the borehole experiment. Therefore, the coverage of  $\delta$  inverted using the proposed method can be wider than that of the mono-component virtual source, although it can be strongly affected by the trade-off with the P-wave velocity. Figs. 15c and 16c support these expectations. This limitation can be mitigated by applying the multistage inversion strategy, which inverts P-wave velocity first and



**Fig. 17.** Depth profiles of the P-wave velocity (a and d),  $\epsilon$  (b and e) and  $\delta$  (c and f) extracted in the true (black line), initial (blue) and inverted model parameters using the mono-component virtual source (green line) and the proposed method (red line) at  $x = 2$  km (a, b and c) and 3.46 km (d, e and f).

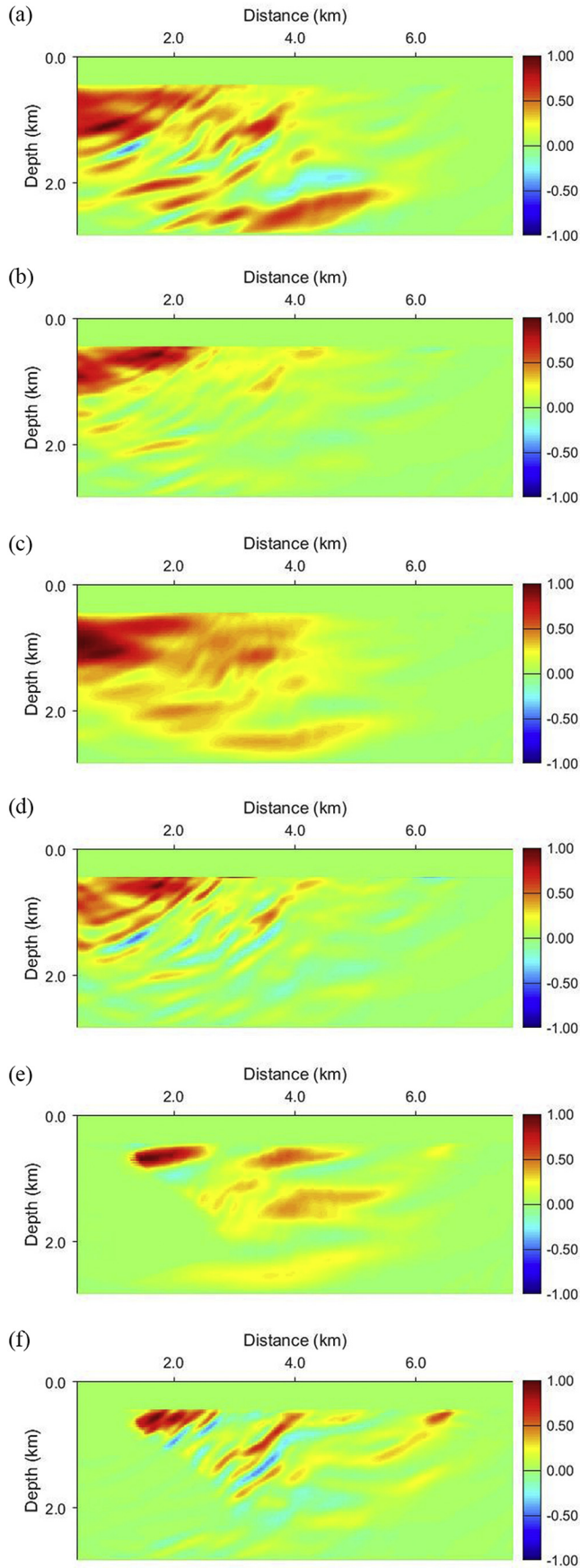
then updates anisotropic parameters. We also extract depth profiles at distances of 2 km and 3.46 km, which include the anomalous areas (Fig. 17). The P-wave velocities obtained by both methods are similar to each other, and the results at a distance of 2 km match the true model better than those at 3.46 km (Fig. 17a and d). From the depth profiles of  $\epsilon$ , we note that the proposed method inverts  $\epsilon$  along the correct direction at the anomalous areas and has a slight trade-off at the bottom (the red arrows in Fig. 17b and e), while  $\epsilon$  obtained using the mono-component virtual source follows the gradient of P-wave velocity. However,  $\delta$  inverted with the proposed method is more affected by the trade-off with the P-wave velocity due to its wider coverage than the gradient of  $\delta$  with the mono-component virtual source (Fig. 17c and f).

In Fig. 18, we show the gradients of  $\epsilon$  and  $\delta$ , which support the FWI results. The gradient of  $\epsilon$  using the conventional method produces a short-wavelength image, which is similar to the structure of the P-wave velocity (Fig. 18a). In contrast, the gradient of  $\epsilon$  using the proposed method is mainly affected by the long-wavelength components and weakly suffers from trade-off with the P-wave velocity in high-wavenumber parts (Fig. 18c). Fig. 18b and d show different coverages of the gradient of  $\delta$  obtained by the conventional virtual source and the proposed method because the partial derivative wavefields of  $\delta$  obtained using the proposed

method can correlate with data at wide incident angles. Fig. 18e and f show the gradients of  $\epsilon$  and  $\delta$  obtained by the coupled equation, respectively. The gradients are distorted around the source location because of the shear-wave artifacts, and consequently, the inversion process does not converge. However, we observe that the gradients are similar to those of our method because both methods use the vector virtual source. From these numerical examples, we conclude that the proposed method can properly update the anisotropic parameters by the correct gradient direction consistent with the elastic scattering theory.

## 5. Discussion

In acoustic FWI for VTI media, the effect of different radiation patterns on the gradient direction depends on the source-receiver configuration and structures of a given model. Fig. 19 shows the theoretical radiation pattern of partial derivative wavefields for  $\delta$  and reflection raypaths at a wide incident angle. In Fig. 19a, we see that for the general surface seismic acquisition with a horizontal layer, the partial derivative wavefield of  $\delta$  with wide incident angles cannot be detected at the receiver because its theoretical radiation pattern dominantly propagates in the vertical directions. On the other hand, the partial derivative wavefield of  $\delta$  for the mono-component virtual source has weak signals



**Fig. 18.** The 1st gradients of the anisotropic parameters  $\epsilon$  (a, c and e) and  $\delta$  (b, d and f) using the mono-component virtual sources (a and b), the proposed method (c and d) and the coupled equation (e and f).

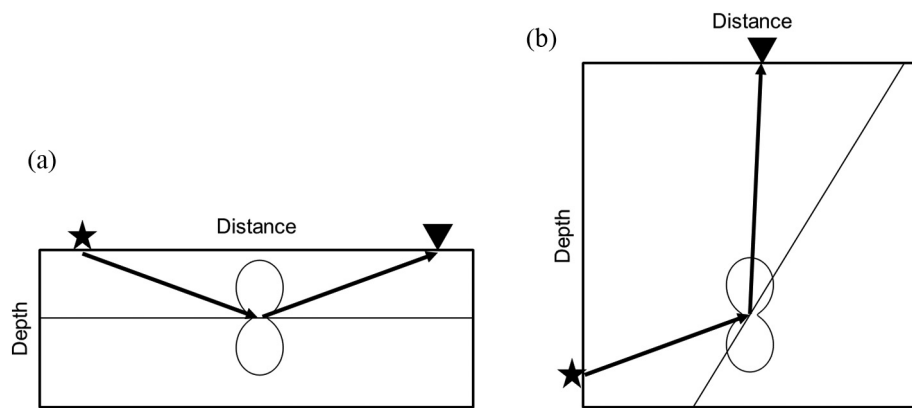
at wide incident angles. In this case, the differences between the gradients of  $\delta$  obtained by the mono-component and vector-based virtual sources are not large; therefore, the effect of the wrong radiation pattern of  $\delta$  on inversion results is not significant for horizontal structures in the surface seismic acquisition. For the borehole seismic acquisition with an inclined layer as shown in Fig. 19b, the reflection raypath arrives vertically at the surface receiver, which matches with the propagation of the partial derivative wavefield of  $\delta$ . Accordingly, the partial derivative wavefield of  $\delta$  can be recorded at the receiver, and it produces wide coverages in the gradient. This phenomenon explains the inversion results in Figs. 15c and 16c.

Although radiation patterns of the partial derivative wavefields also provide information on the dominant wavelength components for model parameters, it is difficult to quantitatively analyze the trade-off between model parameters (Pan et al., 2018b; Podgornova et al., 2018). For real seismic data acquired in complex heterogeneous media, amplitude variations over incident angles can be different. For instance, the radiation pattern of  $\epsilon$  propagates horizontally in a homogeneous background model, but for inclined layers, the dominant scattering angle changes according to the tilt angle, and the propagation direction also varies.

In this paper, to focus on addressing the limitation of the decomposed equation, we adopt only the parameterization including the vertical P-wave velocity and Thomsen parameter, but various parameterizations can be used for acoustic VTI FWI (Gholami et al., 2013). Depending on parameterizations, the radiation patterns of partial derivative wavefields differ, which means that the trade-off between parameters also changes. Therefore, it is important to determine the optimal parameterization according to the accuracy of the initial model and data acquisition. However, with conventional FWI using the mono-component virtual sources, it is difficult to obtain the correct radiation patterns changing according to parameterization. Since the proposed method can be easily applied to other parameterizations, it can be helpful to analyze the influence of parameterization using the decomposed equation with lower computational costs.

## 6. Conclusions

The acoustic-approximation of the elastic wave equations enables us to estimate anisotropic parameters in transversely isotropic media with reduced computational cost. The decomposed equation, which simulates the mono-component pressure fields, advantageously does not generate the shear-wave artifacts and is more efficient than the coupled equations with multicomponent stress or displacement wavefields. In this study, we addressed that the radiation patterns using the mono-component are inconsistent with those obtained from the elastic scattering theory. Because the virtual sources obtained by the mono-component act as explosive sources, the radiation patterns of all the parameters are isotropic, which can lead to the trade-off between model parameters. To overcome this limitation, we proposed using the vector virtual source by converting the pressure wavefield into the displacement components based on the equation of motion. The proposed method yielded radiation patterns and sensitivity kernels similar to those of the coupled equation. We analyzed the effect of the different radiation patterns on inversion through the gradient in the simple syncline model, and the proposed method can locate the syncline structure that conforms to the elastic scattering theory. We applied our method to the modified Marmousi-II model in the borehole seismic experiment. While  $\epsilon$  inverted using the conventional mono-component virtual source suffered from severe trade-off with the P-wave velocity,  $\epsilon$  inverted using the proposed method was less affected by the P-wave velocity and updated more smoothly. In contrast to  $\epsilon$ ,  $\delta$  was updated using our method with wider coverages based on the gradient obtained from various incident angles.



**Fig. 19.** Theoretical radiation patterns of  $\delta$  at an incident angle of  $70^\circ$  (a) for a horizontal layer in surface seismic acquisition and (b) for an inclined layer in a borehole seismic experiment. The black stars and triangles represent the source and receiver positions according to the raypath by Snell's law.

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